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ATTIS: Acoustic Tracking & Target Identification System PROJECT REPORT

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Abstract

The **Acoustic Tracking and Target Identification System (ATTIS)** aims to detect, localise, and eventually navigate toward sound sources in real time. Sound remains a highly reliable clue for locating objects or individuals in visually obscured or disaster-prone environments where other sensing modalities fail. Drawing inspiration from the natural world, where creatures like dolphins, sharks, whales, and bats rely on sound for navigation and communication, and from legendary tales such as that of the warrior Prithviraj Chauhan; who, despite being captured and blinded by Muhammad Ghori, showcased his archery prowess by g only the sound of Ghori's voice to guide his arrow, ultimately killing him.

In this prototype, we developed a stationary sound-localization setup using three microphones and an ESP32 microcontroller. The system captures sound signals via the inbuilt ADC and estimates the direction of the sound source by comparing amplitude and timing differences between microphones. The prototype demonstrates proof of concept for acoustic direction tracking.

Although limited in range and accuracy due to imperfections in microphone sensitivity and cone material & geometry, the prototype validates the feasibility of low-cost acoustic spatial awareness. Future work will focus on precise localisation, mobility integration, enhanced signal processing, and real-time communication to achieve autonomous source tracking and reporting.



Figure 1: ATTIS: as a concept

1 Introduction and Motivation

In many real-world environments, particularly in situations such as:

- Disaster zones,
- Underground mining sites, or
- Military tunnels

visibility is often poor and conditions are highly unpredictable. In such cases, traditional detection methods based on vision or GNSS fail to perform effectively due to the presence of obstacles, dust, and debris. However, sound, being an omnidirectional and naturally occurring signal, can travel through such complex environments and still carry valuable information about its source.

The **Acoustic Tracking and Target Identification System (ATTIS)** has been conceptualized to leverage this property of sound. The system aims to detect the direction of sound, identify its source, and eventually enable a mobile robot to navigate toward it. By focusing on sound-based localization, ATTIS serves as a potential tool for quick and efficient target detection in environments where human should or cannot reach or where optical or GPS-based sensors are unreliable.

The prototype described in this report demonstrates the feasibility of acoustic localization using a triangulated microphone array and the ESP32 microcontroller platform.

1.1 Motivation

The primary motivation behind developing ATTIS is to create an assistive system that can operate effectively in environments where human intervention is either risky or impossible. Sound offers an immediate and reliable means of detecting such events, even when visibility is near zero.

Potential use cases include:

- **Gunshot Direction Detection:** Locating the direction of a gunfire event in real time for military, law enforcement, or anti-poaching operations. The sharp transient nature of a gunshot provides a clear acoustic signature, making direction detection feasible using a distributed microphone array.
- **Search and Rescue:** Detecting cries, tapping, or movement sounds from survivors trapped under rubble during natural disasters such as earthquakes or landslides.
- **Mining Operations:** Monitoring underground acoustic signatures to identify trapped miners or detect signs of collapse or equipment failure.
- **Wildlife Monitoring and Conservation:** Tracking animal calls, detecting illegal hunting activities, or monitoring predator presence in forests.
- **Industrial Monitoring:** Detecting unusual sounds, leaks, or mechanical faults in machinery before a major breakdown occurs.



Figure 2: Application examples for ATTIS

1.2 Objective of the Prototype

The present phase of the project focuses on implementing the sound detection and direction estimation aspects of ATTIS. A stationary setup has been developed to:

1. **Capture sound** from multiple directions using an angular microphone setup.
2. Process and compare signal amplitudes to **estimate the direction** of the source.
3. Provide the **spatial awareness** of the zone in real time.

2 Problem Definition

Sound-based localization involves estimating the direction and position of a sound source using signals captured by multiple microphones. Unlike vision-based systems, acoustic detection does not rely on line of sight and can function effectively in visually obstructed or low-light environments. However, challenges such as background noise, reflections, signal attenuation, and variations in microphone sensitivity introduce significant errors in real-world scenarios. The task, therefore, extends beyond mere sound detection to precise signal handling & processing for accurate direction estimation using limited hardware resources. *In this project, we gave more focus on signal handling by checking reliable data samples, rolling window sizes and a simple pipeline of processing to get spatial awareness*

The **Acoustic Tracking and Target Identification System (ATTIS)** is developed to address this challenge. The system aims to:

- Detect sounds events;

- Distinguish sound sources;
- Determine their source(s), respective directions, and,
- In further iterations, enable autonomous movement toward the source while reporting its position.

Although the present prototype focuses on establishing the feasibility of sound spatial awareness using a stationary setup. It employs an array of three strategically placed *MAX9814* microphones, where each captures sound signals at a gain of 40dB (Auto-gain: Disabled), which are collected and sampled at 8kHz - 16kHz samples per second using the ESP32 microcontroller's inbuilt analogue-to-digital converter (ADC). Direction estimation is achieved by comparing amplitude (mainly) and phase variations between the microphone signals, allowing approximate localisation of the source.

3 System Design and Methodology

3.1 System Overview

The ATTIS system consists of four main modules:

1. **Sound Detection:** Capture ambient sound using an array of three microphones kept at roughly 120°.

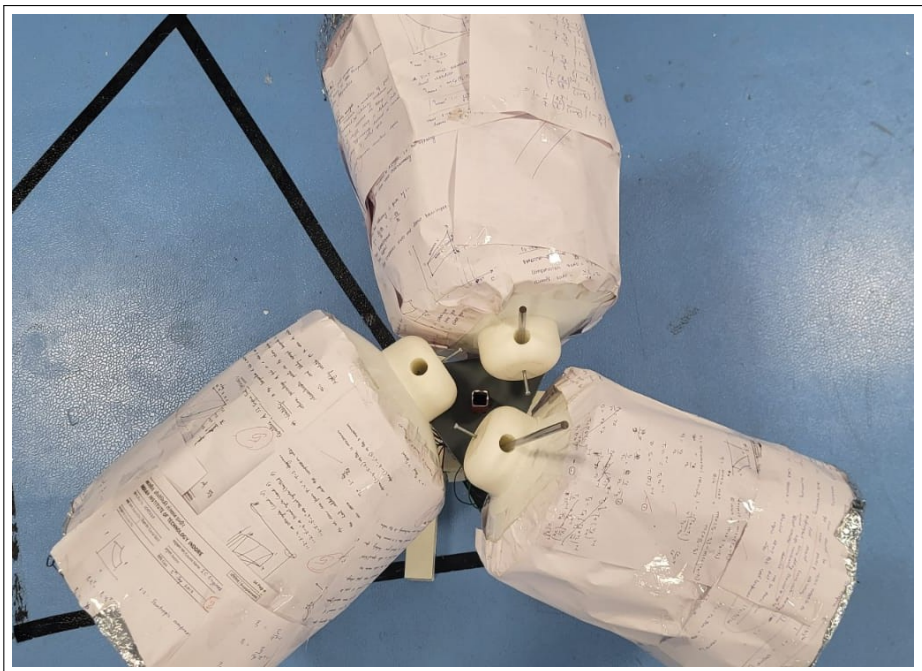


Figure 3: Experimental setup of the sound-receiving structure with mounted microphones.

2. **Signal Processing:** Filter noise and process the sampled digital signals through ESP32 board's ADC to extract significant audio signal features.
3. **Localization:** Estimate source direction based on amplitude (majorly) and phase difference.
4. **Reporting:** Producing a spatially aware sound output with a $\tilde{0.2}$ latency.

3.2 Hardware Setup

- MAX9814 High Performance Microphone AGC Amplifier Module Microphones (3 units) with focusing cones
- ESP32 Wroom development board
- Connecting cables, breadboard, frame, and base structure

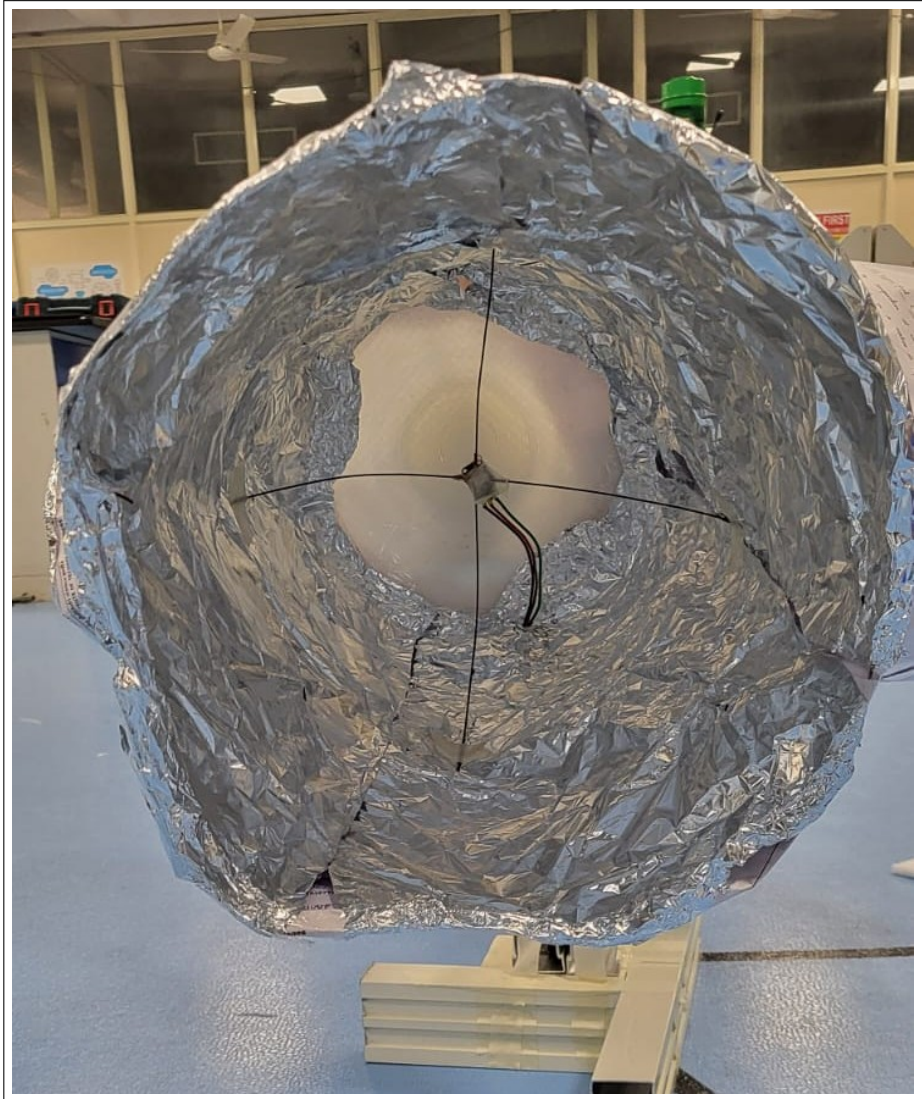


Figure 4: Mounting of microphones at focal point.

3.3 Sound Receiving Structure

The sound-receiving structure was idealised to be designed to concentrate and direct incoming acoustic waves toward the microphones, thereby improving signal strength and directionality. Where each microphone is mounted at the focal point of a saucer-shaped, parabolic dish receiver with a diameter of approximately 25cm and focal length 10cm. One can calculate focal length by: $f = \frac{D^2}{16h}$.

The parabolic geometry enables the receiver to focus incident sound waves effectively, enhancing the signal-to-noise ratio and improving directional sensitivity.

However, due to material constraints within our project timeline, we discovered that the sound was not focused appropriately. Therefore, we covered the saucer-dish with a frustum-like covering made up of a hard material inside and a bit softer outside. Keeping in mind that softer materials will absorb sound vibrations through their deformation energy, while hard, rigid materials (such as Aluminium foil) will have higher reflectivity. This helped us to add more delay to each mic, making the distinction more recognisable and within our range of sampling.

The three receivers are arranged on a rigid triangular stand, forming a stable base for sound localisation. This triangular configuration ensures spatial separation between the microphones by 120 degrees approx., allowing reliable comparison of sound intensity and arrival characteristics across different directions. The structure is currently stationary, but the design allows it to be mounted on a mobile robotic platform in future versions, enabling movement-based tracking and autonomous source localisation in future.

3.4 Signal Processing using ESP32

Sound signals are sampled using ESP32's inbuilt 12-bit ADC. **But it was experimentally observed, the sampling rate need to be significantly less than those bits to get reliable continous real data, and avoid data loss.** The signals are compared in real time, and the relative amplitude difference between microphones is used to estimate the angle of arrival of the sound source. The data is weighted accordingly to be fed in Left and Right outputs of our headphones. *Left = 70% Mic1 + 30% Mic2; Right = 70% Mic3 + 30% Mic2*

3.5 Process Flow Diagram

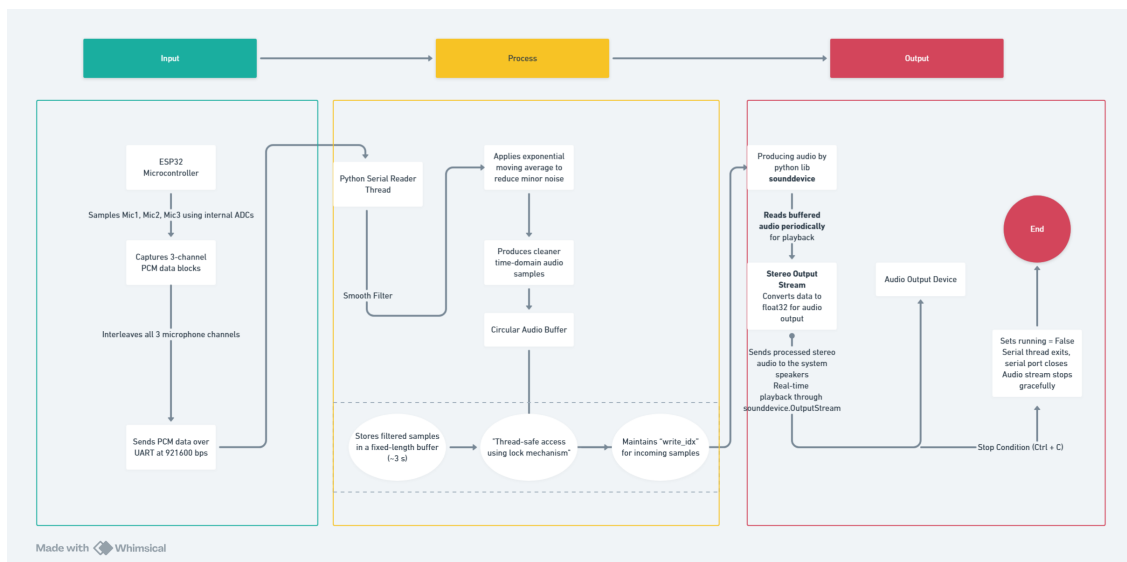


Figure 5: System architecture

4 Results and Observations

- The system successfully detects and differentiates sound direction (left/right).

- Effective range: approximately 1.5–2 meters for clear, distinct sounds.
- Direction accuracy: within $\pm 15\text{--}25^\circ$ for strong sources.
- Performance is still prone to reduction in a noisy environment due to overlapping frequencies.
- However, we hope that when equipped with a real parabolic microphone, our estimated digital signal processing pipeline will improve significantly, allowing for the distinction of sound sources and types.

5 Limitations and Challenges

- Unequal microphone sensitivities and non-ideal frustum-saucer design.
- Noise and resolution limits of ESP32 ADC, limiting synchronisation.
- Stationary platform — lacks motion-based source following and triangulation capabilities due to lack of sound sources.
- Reduced accuracy in complex or reflective environments.

6 Future Work

- Mount the sound array on a mobile robot for autonomous tracking.
- Use digital I²S microphones for synchronized data.
- Implement FFT-Spectrogram-based frequency filtering to isolate target signal sources.
- Add IMU and GPS for real-time localisation and tracking.
- Integrate wireless communication via LoRa or NRF24L01.

7 Conclusion

The ATTIS prototype demonstrates a proof of concept for acoustic direction localization using simple hardware and embedded processing. Despite range and precision limitations, the system establishes the foundation for developing an autonomous sound-tracking robot capable of real-time target identification and reporting.

8 References

1. "A Far-Field Sound Source Localization System for Rescue Robot," *arXiv:2507.21431v1*.
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3. "Applications in Disaster Monitoring," *MDPI, ResearchGate, Wikipedia*.